COURSE NAME: ENVIRONMENTAL SCIENCE
COURSE CODE: NVA-103/203

COURSE OBJECTIVES:

1. To create and disseminate knowledge to the students about environmental problems atlocal, regional and global scale.

1. To understand the basic concepts of Environments and its components along with their interactions through study of Ecology, Biodiversity, and different types of pollution.

3. To sensitize students towards environmental concerns, issues, and impacts of climatechange and related mitigation strategies.

4. To make the students to apply their knowledge for efficient environmental decision-making, management and sustainable development.

5. To prepare students for successful career in environmental departments, research institutes, industries, consultancy and NGOs, etc.

Unit I

Introduction to Environmental Science: Environmental segments and its scope, Natural resources and its conservation, Introduction to Green chemistry, Goals of Green chemistry, Industrial application of Green chemistry, Environmental law's: Environmental Protection Act-1986, The Water Act-1974, The Air Act-1981, Wildlife Protection Act-1972, Basics of Environmental Impact Assessment, Environmental Audit

Unit II

Ecosystem: Ecosystem and its basic concept, Structure and function of an ecosystem, Types of ecosystem, Food chain, Food web, Material Cycles (Carbon, Nitrogen, phosphorus and Sulphur), Ecological succession, Ecological pyramids, Formation of Soil, Composition of Soil

Unit III

Biodiversity: Biodiversity and its conservation, Ex-situ and In-situ conservation (National Parks, Biosphere Reserves, Wildlife Sanctuaries, Sacred grooves), Types of Biodiversity, Hotspots of Biodiversity, Threat to biodiversity, Biodiversity conservation plan (Project Tiger, Project Elephant, Project Crocodile, Project Rhino)

Unit IV

Pollution: Environmental Pollution, Air pollution (Primary and secondary Pollutants), Water Pollution, Soil Pollution, Noise Pollution, Radioactive Pollution, Sources of Pollution, Impact of Pollution on

Environment, Methods of monitoring and control of Environmental Pollution, Air Quality Index, Waste Water treatment plants.

Unit V

Social issues: Sustainable development and its goals, Climate change, Global warming, Greenhouse effect, Population Growth, Acid rain, Ozone layer depletion, Biomagnification, Solid Waste Management, Narmada bachao andolan, Chipko Movement, Appiko movement, Need for Public Awareness on Environmental Issues.

Course Outcomes:

After completing this course, a student will be able to-

CO1: Gain knowledge about environment and ecosystem.

CO2: Students will learn about natural resource, its importance and environmental impacts of human activities on natural resource.

CO3: Gain knowledge about the conservation of biodiversity and its importance.

CO4: Aware students about problems of environmental pollution, its impact on human and ecosystem and control measures.

CO5: Students will be able to evaluate contribution of Green Houses Gases in Globalwarming and thereby bringing Change in Climate.

Suggested Readings:

Gilbert M. Master and Wentdell P. Ella (2018) Introduction to Environmental Engineering and Science. Pearson India Education Services Pvt. Ltd. Noida UP India.

S.S. Dara (2009) A text book of Environmental Chemistry and Pollution Control. S. Chand & Company Ltd. New Delhi India.

R.J. Ranjit Daniels and Jagdish Krishnaswamy (2013) Environmental Studies.John Wiley & Sons, 111 River Street, Hoboken, NJ 07030, USA.

Arnab Kumar De (2017). Environmental Chemistry. New Age International Pvt.Ltd. Publishers, New Delhi, India.

Erach Bharucha (2019) Textbook of Environmental Studies. E-book epubs@textsoft.in

J.S. Singh, S.P. Singh, S.R. Gupta (2014) Ecology, Environmental Science andConservation. S. Chand & Company Ltd. New Delhi India.